EXCEL ACADEMY

Metals and Non-Metals - Class 10 Chemistry (CBSE)

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1. Introduction

Metals and non-metals are two major categories of elements in the periodic table. They differ significantly in their physical and chemical properties.

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2. Physical Properties of Metals

2.1 Metallic Lustre

- Metals have a shiny appearance called metallic lustre.
- Freshly cut metal surfaces are bright and shiny.
- Examples: Gold, silver, and copper have excellent lustre.
- Metals lose their shine when exposed to air due to formation of oxide layers.

2.2 Hardness

- Most metals are hard and cannot be cut easily with a knife.
- Exception: Sodium and potassium are soft metals that can be cut with a knife.
- Hardest metal is chromium.
- Metals like lead and mercury are relatively soft.

2.3 Malleability

- Malleability is the property of metals to be beaten into thin sheets.
- Gold is the most malleable metal.
- One gram of gold can be beaten into a thin sheet of area 1 square meter.
- Silver and aluminium are also highly malleable.
- Malleability allows metals to be shaped into various forms.

2.4 Ductility

- Ductility is the property of metals to be drawn into thin wires.
- Gold is the most ductile metal.
- One gram of gold can be drawn into a wire of about 2 kilometers length.
- Copper, aluminium, and silver are also highly ductile.
- This property makes metals useful for making electrical wires.

2.5 Conductivity of Heat and Electricity

- All metals are good conductors of heat and electricity.
- Silver is the best conductor of heat and electricity.
- Copper and aluminium are also excellent conductors.
- Metals conduct electricity because they have free electrons.
- Lead and mercury are comparatively poor conductors.

2.6 Sonority

- Metals produce a ringing sound when struck hard.
- This property is called sonority.
- Metals are called sonorous.
- This property is used in making bells and musical instruments.

2.7 Density and Melting Point

- Most metals have high density.
- Sodium and potassium are exceptions with low density (they can float on water).
- Most metals have high melting and boiling points.
- Exception: Gallium and cesium have very low melting points (they can melt in hand).
- Mercury is liquid at room temperature.

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3. Physical Properties of Non-Metals

3.1 Appearance

- Non-metals generally lack metallic lustre.
- They are dull in appearance.
- Exception: Iodine and graphite (an allotrope of carbon) have lustre.

3.2 Hardness

- Non-metals are generally soft.
- Exception: Diamond (an allotrope of carbon) is the hardest natural substance.

3.3 Malleability and Ductility

- Non-metals are non-malleable and non-ductile.
- They are brittle in solid state.
- When force is applied, they break into pieces.

3.4 Conductivity

- Non-metals are poor conductors of heat and electricity.
- They are called insulators.
- Exception: Graphite is a good conductor of electricity because it has free electrons.

3.5 Sonority

- Non-metals are not sonorous.
- They do not produce ringing sound when struck.

3.6 Density and Melting Point

- Non-metals generally have low density.
- They have low melting and boiling points.
- Exception: Diamond and graphite have very high melting points.

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4. Chemical Properties of Metals

4.1 Reaction with Oxygen

- Metals react with oxygen to form metal oxides.
- **General Equation:** Metal + Oxygen → Metal Oxide

Examples:

- $4Na + O_2 \rightarrow 2Na_2O$ (Sodium oxide)
- $2Mg + O_2 \rightarrow 2MgO$ (Magnesium oxide)
- $4Al + 3O_2 \rightarrow 2Al_2O_3$ (Aluminium oxide)
- $2Cu + O_2 \rightarrow 2CuO$ (Copper oxide)

Nature of Metal Oxides:

- Metal oxides are basic in nature.
- They turn red litmus paper blue.
- Metal oxides react with acids to form salt and water.
- Example: $CuO + 2HCl \rightarrow CuCl_2 + H_2O$

Amphoteric Oxides:

- Some metal oxides react with both acids and bases.
- These are called amphoteric oxides.
- Examples: Aluminium oxide (Al_2O_3) and Zinc oxide (ZnO).
- $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$ (reacts with acid)
- $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$ (reacts with base)

Anodising:

- Anodising is a process of forming a thick oxide layer on aluminium.
- It protects aluminium from corrosion.
- The layer can be dyed to give attractive colors.

4.2 Reaction with Water

- Metals react with water to form metal hydroxide and hydrogen gas.
- **General Equation:** Metal + Water → Metal Hydroxide + Hydrogen

Different metals react differently with water:

a) Metals that react with cold water:

- Sodium and potassium react vigorously with cold water.
- $2Na + 2H_2O \rightarrow 2NaOH + H_2$ (Vigorous reaction)
- $2K + 2H_2O \rightarrow 2KOH + H_2$ (Explosive reaction)
- Calcium reacts less vigorously with cold water.
- $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$

b) Metals that react with hot water or steam:

- Magnesium reacts slowly with hot water but vigorously with steam.
- $Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$ (hot water)
- $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$ (steam - iron)
- Aluminium and zinc also react with steam.

c) Metals that do not react with water:

- Gold, silver, copper, and platinum do not react with water at all.
- Iron reacts with steam but not with cold or hot water under normal conditions.

4.3 Reaction with Acids

- Metals react with acids to form salt and hydrogen gas.
- **General Equation:** Metal + Acid → Salt + Hydrogen gas

Examples:

- $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
- $Mg + H_2SO_4 \rightarrow MgSO_4 + H_2$
- $Fe + 2HCl \rightarrow FeCl_2 + H_2$

Important Points:

- Hydrogen gas burns with a pop sound.
- Not all metals react with all acids.
- Copper does not react with dilute HCl or H_2SO_4.
- Metals like sodium and potassium react explosively with acids.
- Nitric acid (HNO_3) is an oxidizing acid and does not produce hydrogen with metals (except Mg and Mn).
- Aqua regia (mixture of concentrated HCl and HNO_3 in 3:1 ratio) can dissolve gold and platinum.

4.4 Reaction with Bases

- Not all metals react with bases.
- Only some metals like aluminium, zinc, and tin react with bases.
- They form salt and hydrogen gas.

Examples:

- $2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$ (Sodium aluminate)
- $Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$ (Sodium zincate)

4.5 Displacement Reactions

- A more reactive metal can displace a less reactive metal from its salt solution.
- This is called displacement reaction.

Examples:

- $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ (Zinc displaces copper)
- $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (Iron displaces copper)
- $Mg + CuSO_4 \rightarrow MgSO_4 + Cu$ (Magnesium displaces copper)

Observations:

- The blue color of copper sulphate solution fades.
- Reddish-brown copper metal deposits on the zinc/iron/magnesium.

Note: Copper cannot displace zinc from zinc sulphate because copper is less reactive than zinc.

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5. Chemical Properties of Non-Metals

5.1 Reaction with Oxygen

- Non-metals react with oxygen to form non-metallic oxides.
- **General Equation:** Non-metal + Oxygen → Non-metallic oxide

Examples:

- $C + O_2 \rightarrow CO_2$ (Carbon dioxide)
- $S + O_2 \rightarrow SO_2$ (Sulphur dioxide)
- $4P + 5O_2 \rightarrow 2P_2O_5$ (Phosphorus pentoxide)

Nature of Non-metallic Oxides:

- Non-metallic oxides are acidic in nature.
- They turn blue litmus paper red.
- They react with bases to form salt and water.
- Example: $CO_2 + 2NaOH \rightarrow Na_2CO_3 + H_2O$

5.2 Reaction with Water

- Most non-metals do not react with water.
- Some non-metallic oxides dissolve in water to form acids.

Examples:

- $CO_2 + H_2O \rightarrow H_2CO_3$ (Carbonic acid)
- $SO_2 + H_2O \rightarrow H_2SO_3$ (Sulphurous acid)

5.3 Reaction with Acids

- Non-metals generally do not react with acids.
- This is because non-metals cannot lose electrons easily.

5.4 Reaction with Bases

- Some non-metals react with bases.
- Example: Chlorine reacts with sodium hydroxide.
- $Cl_2 + 2NaOH \rightarrow NaCl + NaOCl + H_2O$

5.5 Displacement Reactions

- More reactive non-metals can displace less reactive non-metals from their salt solutions.
- Example: $Cl_2 + 2KBr \rightarrow 2KCl + Br_2$ (Chlorine displaces bromine)

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6. Reactivity Series

The reactivity series is an arrangement of metals in decreasing order of their reactivity.

Reactivity Series (from most reactive to least reactive):

Most Reactive:

- Potassium (K)
- Sodium (Na)
- Calcium (Ca)
- Magnesium (Mg)
- Aluminium (Al)
- Zinc (Zn)
- Iron (Fe)
- Lead (Pb)
- **Hydrogen (H)** - Not a metal but included for reference
- Copper (Cu)
- Mercury (Hg)
- Silver (Ag)
- Gold (Au)
- Platinum (Pt) **Least Reactive**

Uses of Reactivity Series:

1. Metals above hydrogen in the series can displace hydrogen from acids.
2. Metals below hydrogen cannot displace hydrogen from acids.
3. A more reactive metal can displace a less reactive metal from its salt solution.
4. Metals at the top are more reactive and readily lose electrons.
5. Metals at the bottom are less reactive and do not lose electrons easily.

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7. How Metals React - Ionic Nature

7.1 Formation of Ionic Compounds

Why metals form ionic compounds:

- Metals have 1, 2, or 3 electrons in their outermost shell.
- They lose these electrons to achieve stable electronic configuration.
- By losing electrons, metals form positive ions (cations).

Example: Sodium Chloride (NaCl) Formation

- Sodium (Na) has electronic configuration: 2, 8, 1
- Sodium loses 1 electron to form Na^+ ion with configuration 2, 8
- Chlorine (Cl) has electronic configuration: 2, 8, 7
- Chlorine gains 1 electron to form Cl^- ion with configuration 2, 8, 8
- Na^+ and Cl^- ions attract each other to form NaCl (ionic bond)

Example: Magnesium Oxide (MgO) Formation

- Magnesium (Mg) has configuration: 2, 8, 2
- Magnesium loses 2 electrons to form Mg^{2+} with configuration 2, 8
- Oxygen (O) has configuration: 2, 6
- Oxygen gains 2 electrons to form O^{2-} with configuration 2, 8
- Mg^{2+} and O^{2-} attract to form MgO

7.2 Properties of Ionic Compounds

1. **Physical Nature:** Ionic compounds are solid and hard due to strong electrostatic force of attraction.
2. **Melting and Boiling Points:** They have high melting and boiling points.
3. **Solubility:** They are generally soluble in water but insoluble in organic solvents like kerosene, petrol.
4. **Electrical Conductivity:**
    - They do not conduct electricity in solid state.
    - They conduct electricity in molten state or aqueous solution.
    - This is because ions become free to move in molten or aqueous state.

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8. Occurrence of Metals

8.1 Forms of Metal Occurrence

a) Free State (Native State):

- Metals which are least reactive occur in free state.
- Examples: Gold, silver, platinum, and copper.
- These metals are found at the bottom of reactivity series.

b) Combined State:

- Most metals are reactive and occur in combined form.
- They are found as compounds with oxygen, sulphur, halogens, etc.
- These compounds are called minerals.

8.2 Important Terms

Minerals:

- Naturally occurring compounds of metals found in the earth's crust are called minerals.
- Not all minerals are suitable for extraction of metals.

Ores:

- Minerals from which metals can be extracted profitably and conveniently are called ores.
- All ores are minerals but all minerals are not ores.
- Example: Clay ($Al_2O_3.2H_2O$) and Bauxite ($Al_2O_3.2H_2O$) both contain aluminium. But bauxite is an ore because aluminium can be extracted profitably from it.

Gangue:

- Ores mined from earth contain impurities like sand, clay, rocks, etc.
- These impurities are called gangue or matrix.
- Gangue must be removed before extraction of metal.

8.3 Common Ores

Metal	Ore Name	Chemical Formula
Aluminium	Bauxite	$Al_2O_3.2H_2O$
Iron	Haematite	Fe_2O_3
Iron	Magnetite	Fe_3O_4
Copper	Copper pyrites	$CuFeS_2$
Copper	Malachite	$CuCO_3.Cu(OH)_2$
Zinc	Zinc blende	ZnS
Zinc	Calamine	$ZnCO_3$
Lead	Galena	PbS
Sodium	Rock salt	$NaCl$

9. Extraction of Metals (Metallurgy)

The process of extracting metals from their ores is called metallurgy. It involves several steps.

9.1 Enrichment of Ores (Concentration)

The process of removing gangue from ore is called concentration or enrichment.

Methods of Concentration:

a) Hand Picking:

- Used when ore and gangue have large difference in appearance.
- Simply picked by hand or washed with water.

b) Hydraulic Washing (Gravity Separation):

- Based on difference in densities of ore and gangue.
- Ore is heavier and settles down while gangue is lighter and washed away with water.

c) Magnetic Separation:

- Used when ore is magnetic.
- Ore is separated from gangue using magnets.
- Example: Magnetite (Fe_3O_4) can be separated using this method.

d) Froth Flotation:

- Used for sulphide ores.
- Ore particles are wetted by oil and gangue by water.
- When mixture is agitated with water, ore particles stick to oil froth and float.
- Gangue settles down at bottom.

9.2 Conversion of Concentrated Ore to Metal Oxide

a) Calcination:

- Heating of carbonate ore in limited supply of air below its melting point.
- Carbonate ore is converted to oxide ore.
- Volatile impurities and moisture are removed.
- Example: $ZnCO_3 \rightarrow ZnO + CO_2$ (Heat)

b) Roasting:

- Heating of sulphide ore in excess of air below its melting point.
- Sulphide ore is converted to oxide ore.
- Example: $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$ (Heat)

9.3 Reduction of Metal Oxides to Metals

a) Reduction using Carbon:

- Metal oxides of less reactive metals are reduced by heating with carbon (coke).
- Carbon acts as reducing agent.

Examples:

- $ZnO + C \rightarrow Zn + CO$
- $Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$
- $SnO_2 + 2C \rightarrow Sn + 2CO$

b) Reduction by Aluminium (Thermit Process):

- Metal oxides of moderately reactive metals can be reduced using aluminium.
- Aluminium has more affinity for oxygen than these metals.
- This process is called thermit reaction.

Example:

- $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 + Heat$
- This reaction produces a lot of heat.
- Used for welding railway tracks and broken machinery parts.

c) Electrolytic Reduction:

- Highly reactive metals like sodium, potassium, calcium, magnesium, aluminium cannot be reduced by carbon or aluminium.
- They have more affinity for oxygen than carbon.
- These metals are obtained by electrolytic reduction of their molten ores.
- Electric current is passed through molten ore.
- Metal is deposited at cathode.

Example: Extraction of Aluminium from Bauxite

- Bauxite ($Al_2O_3.2H_2O$) is purified to get pure aluminium oxide (Al_2O_3 or Alumina).
- Alumina has very high melting point.
- Cryolite (Na_3AlF_6) and fluorspar (CaF_2) are added to lower melting point.
- Electrolysis is carried out.
- At cathode: $Al^{3+} + 3e^- \rightarrow Al$
- At anode: $2O^{2-} \rightarrow O_2 + 4e^-$
- Oxygen reacts with carbon anode to form CO_2.

9.4 Refining of Metals

The extracted metal contains impurities. The process of purifying impure metal is called refining.

Methods of Refining:

a) Electrolytic Refining:

- Used for refining of copper, zinc, gold, silver, etc.

- Impure metal is made anode.
- Thin strip of pure metal is made cathode.
- Salt solution of metal is used as electrolyte.
- On passing current, pure metal from anode dissolves and deposits at cathode.
- Impurities settle down below anode as anode mud.

Example: Refining of Copper

- Anode: Impure copper
- Cathode: Pure copper strip
- Electrolyte: Copper sulphate solution with dil. H_2SO_4
- At anode: $Cu \rightarrow Cu^{2+} + 2e^-$
- At cathode: $Cu^{2+} + 2e^- \rightarrow Cu$

b) Liquation:

- Used for metals with low melting point like tin, lead, bismuth.
- Impure metal is heated on sloping furnace.
- Pure metal melts and flows down leaving impurities behind.

c) Distillation:

- Used for metals with low boiling point like mercury, zinc.
- Impure metal is heated to form vapours.
- Vapours are cooled to get pure metal.

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10. Corrosion

10.1 Definition and Examples

- Corrosion is the process of slow eating away of metal due to attack of atmospheric gases and moisture on its surface.
- It results in formation of compounds like oxides, carbonates, sulphides, etc.

Examples:

- Rusting of iron: Formation of reddish-brown coating (hydrated ferric oxide - $Fe_2O_3.xH_2O$)
- Black coating on silver: Formation of silver sulphide (Ag_2S)
- Green coating on copper: Formation of copper carbonate [$CuCO_3.Cu(OH)_2$]

10.2 Conditions for Rusting

- Rusting of iron requires both oxygen and water.
- In dry air, iron does not rust.
- In oxygen-free water, iron does not rust.

10.3 Harmful Effects

- Corrosion weakens the metal structure.
- It causes economic loss as corroded metal articles become useless.
- Large amounts of money are spent in replacing damaged iron articles.

10.4 Prevention of Corrosion

a) Painting:

- Coating metal surface with paint prevents contact with air and moisture.
- Used for iron gates, bridges, railings, etc.

b) Oiling and Greasing:

- Applying oil or grease prevents rusting.
- Used for machine parts, tools, etc.

c) Galvanization:

- Coating iron with a layer of zinc.
- Zinc is more reactive than iron and gets corroded first.
- Zinc protects iron from corrosion.
- Used for water pipes, roofing sheets, buckets, etc.

d) Electroplating:

- Depositing a layer of non-corrosive metal on reactive metal by electrolysis.
- Examples: Chromium plating on iron, tin plating on iron, gold plating on silver.

e) Alloying:

- Mixing two or more metals to form alloy.
- Alloys are more resistant to corrosion.
- Example: Stainless steel (iron + chromium + nickel) does not rust.

f) Anodising:

- For aluminium: forming thick oxide layer on surface.
- Protects from further corrosion.

11. Alloys

11.1 Definition

- An alloy is a homogeneous mixture of two or more metals, or a metal and a non-metal.
- Alloys are prepared by melting the constituent metals together and cooling.

11.2 Properties of Alloys

- Alloys are stronger than pure metals.
- They have lower melting points than pure metals.
- They are more resistant to corrosion.
- They have different properties from constituent metals.
- They are harder than pure metals.

11.3 Common Alloys and Their Uses

Alloy	Composition	Uses
Brass	Copper + Zinc	Decorative items, musical instruments, utensils
Bronze	Copper + Tin	Statues, medals, coins
Solder	Lead + Tin	Welding electrical wires
Stainless Steel	Iron + Chromium + Nickel	Utensils, cutlery, surgical instruments
Steel	Iron + Carbon (0.1% to 1.5%)	Construction, bridges, machines
Duralumin	Aluminium + Copper + Magnesium + Manganese	Aircraft bodies, pressure cookers
Amalgam	Mercury + any metal	Dental fillings (Mercury + Silver + Tin)
German Silver	Copper + Zinc + Nickel	Decorative items, resistance wires

11.4 Why Alloys are Preferred

1. **Pure gold (24 carat) is very soft** and cannot be used for making jewelry. It is alloyed with copper or silver to make it hard. 22 carat gold contains 22 parts gold and 2 parts copper.
2. **Pure iron is soft** and easily gets stretched. Mixing carbon makes it hard and strong (steel).
3. **Stainless steel does not rust** unlike pure iron, making it ideal for utensils and surgical instruments.
4. **Solder has low melting point**, making it suitable for welding purposes.
5. **Alloys are more resistant to corrosion** than pure metals.

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12. Important Chemical Equations (Quick Reference)

Reaction with Oxygen:

- $2Mg + O_2 \rightarrow 2MgO$
- $4Al + 3O_2 \rightarrow 2Al_2O_3$
- $2Cu + O_2 \rightarrow 2CuO$

Reaction with Water:

- $2Na + 2H_2O \rightarrow 2NaOH + H_2\uparrow$
- $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2\uparrow$
- $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2\uparrow$ (steam)

Reaction with Acids:

- $Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$
- $Mg + H_2SO_4 \rightarrow MgSO_4 + H_2\uparrow$
- $2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2\uparrow$

Reaction with Bases:

- $2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2\uparrow$
- $Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2\uparrow$

Displacement Reactions:

- $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
- $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
- $Mg + CuSO_4 \rightarrow MgSO_4 + Cu$

Thermit Reaction:

- $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 + Heat$

Calcination:

- $ZnCO_3 \rightarrow ZnO + CO_2$ (Heat)
- $CaCO_3 \rightarrow CaO + CO_2$ (Heat)

Roasting:

- $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$ (Heat)
- $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$ (Heat)

Reduction:

- $ZnO + C \rightarrow Zn + CO$
- $Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$
- $CuO + C \rightarrow Cu + CO$

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14. Exam-Oriented Important Points

14.1 One-Mark Questions Topics:

- Definition of malleability, ductility, sonority
- Name the most malleable and ductile metal
- Name metals that can be cut with knife
- Name the best conductor of heat and electricity
- What is an alloy?
- Name the metal that is liquid at room temperature
- What is anodising?
- Chemical formula of rust
- Name the gas produced when metal reacts with acid

14.2 Two-Mark Questions Topics:

- Difference between metals and non-metals (any 4 points)
- Why is sodium kept immersed in kerosene?
- What are amphoteric oxides? Give two examples
- Define: ore, mineral, gangue
- Write two methods of preventing rusting
- Why do ionic compounds conduct electricity in molten state?
- What is thermit reaction? Write equation
- Name two methods of concentration of ores

14.3 Three-Mark Questions Topics:

- Write three differences between ionic and covalent compounds
- Explain why gold and platinum are found in free state
- How is copper refined? Explain with diagram
- Explain the process of formation of ionic compound (NaCl or MgO)
- Why are alloys preferred over pure metals? Give three reasons
- Explain reactivity series and its importance
- Distinguish between calcination and roasting

14.4 Five-Mark Questions Topics:

- Explain the steps involved in extraction of metals from ores
- Describe the extraction of aluminium from bauxite with equation
- Write all the chemical properties of metals with equations
- Explain corrosion. Why does iron rust? How can it be prevented?
- Describe three methods of concentration of ores
- Explain how metals react with oxygen, water, and acids with equations

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15. Important Definitions (Learn by Heart)

1. **Metals:** Elements that lose electrons to form positive ions and have properties like lustre, malleability, ductility, and conductivity.
2. **Non-metals:** Elements that gain electrons to form negative ions and lack metallic properties.
3. **Malleability:** The property of metals to be beaten into thin sheets.
4. **Ductility:** The property of metals to be drawn into thin wires.
5. **Sonority:** The property of metals to produce ringing sound when struck.
6. **Metallic lustre:** The shiny appearance of metals.
7. **Mineral:** Naturally occurring compounds of elements found in earth's crust are called minerals.

8. **Ore:** Minerals from which metals can be extracted profitably and conveniently.
9. **Gangue:** The impurities like sand, clay, and rocks present in ores.
10. **Metallurgy:** The process of extracting metals from their ores and refining them.
11. **Concentration/Enrichment:** The process of removing gangue from ore.
12. **Calcination:** Heating of carbonate ore in limited supply of air below melting point to convert it into oxide.
13. **Roasting:** Heating of sulphide ore in excess of air below melting point to convert it into oxide.
14. **Reduction:** The process of removing oxygen from metal oxide to obtain metal.
15. **Refining:** The process of purifying impure metal.
16. **Alloy:** A homogeneous mixture of two or more metals, or a metal and a non-metal.
17. **Corrosion:** The process of slow eating away of metal surface due to attack of atmospheric gases and moisture.
18. **Rusting:** Corrosion of iron to form hydrated ferric oxide (rust).
19. **Galvanization:** The process of coating iron with a layer of zinc to prevent rusting.
20. **Anodising:** The process of forming thick oxide layer on aluminium to protect it from corrosion.
21. **Amphoteric oxides:** Metal oxides that react with both acids and bases.
22. **Ionic compounds:** Compounds formed by transfer of electrons between metal and non-metal atoms.
23. **Reactivity series:** Arrangement of metals in decreasing order of their reactivity.
24. **Aqua regia:** Mixture of concentrated HCl and HNO_3 in 3:1 ratio that can dissolve gold and platinum.
25. **Thermit reaction:** Reduction of metal oxide using aluminium as reducing agent, producing lot of heat.

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16. Common Mistakes to Avoid

1. **Don't confuse calcination and roasting** - Remember: Calcination is for carbonates (limited air), roasting is for sulphides (excess air).
2. **Nitric acid doesn't produce hydrogen** with most metals - It's an oxidizing acid.
3. **Not all minerals are ores** - But all ores are minerals.
4. **Hydrogen is not a metal** - It's included in reactivity series only for reference.
5. **Metals don't lose their lustre** - They appear dull due to formation of oxide layer on surface.
6. **Ionic compounds conduct electricity** only in molten state or aqueous solution, NOT in solid state.
7. **Copper doesn't react** with dilute HCl or H_2SO_4.
8. **Sodium and potassium are stored in kerosene** - Not in water (they react vigorously with water).
9. **Gold is most malleable and ductile** - Not silver (though silver is best conductor).
10. **Diamond is an allotrope of carbon** - It's a non-metal but hardest natural substance.

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18. Practice Questions for Exam

Very Short Answer (1 Mark):

1. Name the most abundant metal in earth's crust.
2. Which metal is used for galvanization?
3. What is the composition of brass?
4. Name one metal that reacts with cold water.
5. What type of oxides are formed by non-metals?

Short Answer I (2 Marks):

1. Why do gold and silver not corrode in moist air?
2. Give reasons why platinum and gold are used to make jewellery.
3. Why is aluminium used for making electrical wires?
4. What is meant by refining of metals?
5. Name two metals which are both malleable and ductile.

Short Answer II (3 Marks):

1. State reasons for the following: a) Carbonate and sulphide ores are converted to oxides. b) Metals like Na, K, Ca, Mg are obtained by electrolytic reduction.
2. Differentiate between metal and non-metal on the basis of their chemical properties.

3. What is meant by corrosion? How can we prevent it?

Long Answer (5 Marks):

1. Describe the method of extraction of copper from copper pyrites with equations.
2. What are amphoteric oxides? Choose amphoteric oxides from: Na_2O, ZnO, Al_2O_3, CO_2, H_2O. Write chemical equations to justify your answer.
3. Explain giving reasons: a) Reactivity of Al decreases if dipped in HNO_3 b) Carbon cannot reduce oxides of Na, Mg c) Metals like Na, K are stored under kerosene d) Marble statues get corroded by acid rain

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19. Exam Tips

1. **Always write balanced chemical equations** - Half marks may be deducted for unbalanced equations.
2. **Draw neat diagrams** for electrolytic refining, thermit reaction, etc.
3. **Underline important terms** like metallurgy, calcination, roasting, etc. in answers.
4. **Use proper chemical formulas** - Don't write "rust" instead of $Fe_2O_3.xH_2O$.
5. **Mention observations** in activity-based questions (color change, gas evolution, etc.).
6. **Learn all chemical equations** given in NCERT textbook.
7. **Practice writing differences** in tabular form for better presentation.
8. **Remember exceptions** - Like graphite conducts electricity, diamond is hardest non-metal, sodium-potassium are soft metals.
9. **Use scientific terms** - Don't write "shiny" instead of "metallic lustre".
10. **Time management** - Allocate time according to marks: 1 mark = 1 minute approximately.
